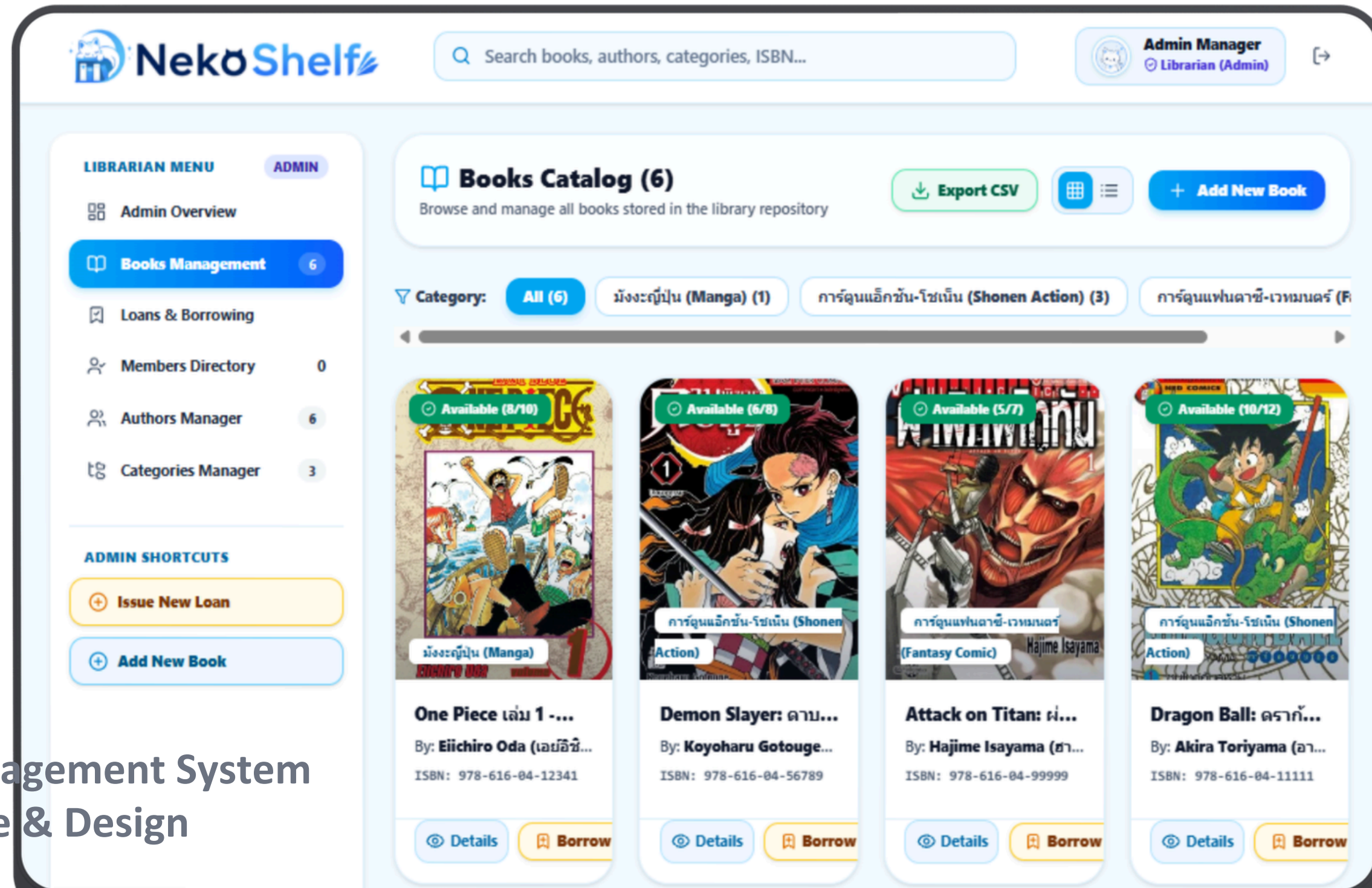




Library & Manga Management System  
Database Architecture & Design

# Project Overview

NekoShelf is a modern web application designed to manage library inventories and manga collections. It supports two user roles: Librarians (Admin) for inventory management and Members (Readers) for browsing and borrowing books.

## Core Responsibilities:

- Manage book and manga catalog with ISBN tracking
- Track borrowing and return transactions
- Monitor member accounts and borrowing history
- Generate reports and analytics

# Key Features

## Book Catalog

Search, filter, and browse library inventory by title, author, category, or ISBN

## Loans Management

Issue, track, and process book returns with automated inventory updates

## Member Directory

Maintain member profiles with borrowing history and account status

## Authors & Categories

Manage author profiles and organize books into categories

## Analytics Dashboard

View statistics on popular books, member activity, and overdue items

## Export Reports

Export loan data to CSV for analysis and reporting

# Database Architecture

*NekoShelf uses Supabase Cloud (PostgreSQL) with 6 core tables*

-  authors – Writer/illustrator profiles
-  categories – Book classification
-  books – Catalog inventory
-  members – Library member accounts
-  loans – Borrowing transactions
-  users – System user permissions

# Database Schema (1/3)

## AUTHORS

id (BIGINT, PK)

name (TEXT)

bio (TEXT)

image (TEXT)

created\_at (TIMESTAMPTZ)

## CATEGORIES

id (BIGINT, PK)

name (TEXT, Unique)

description (TEXT)

created\_at (TIMESTAMPTZ)

# Database Schema (2/3)

## BOOKS

id (BIGINT, PK) • title (TEXT)

author\_id (BIGINT, FK) • category\_id (BIGINT, FK)

isbn (TEXT) • published\_year (INT)

total\_copies (INT) • available\_copies (INT)

cover\_image (TEXT) • description (TEXT)

created\_at (TIMESTAMPTZ)

## MEMBERS

id (BIGINT, PK) • member\_code (TEXT, Unique)

name (TEXT) • email (TEXT) • phone (TEXT)

created\_at (TIMESTAMPTZ)

# Database Schema (3/3)

## LOANS

id (BIGINT, PK)

book\_id (BIGINT, FK) • member\_id (BIGINT, FK)

borrow\_date (DATE) • due\_date (DATE)

return\_date (DATE, Nullable) • created\_at (TIMESTAMPTZ)

## USERS

id (BIGINT, PK) • username (TEXT, Unique) • email (TEXT, Unique)

role (TEXT: 'admin' or 'user') • name (TEXT)

created\_at (TIMESTAMPTZ)

# Technology Stack

## Frontend

React 19, Vite 6, Tailwind CSS v4, React Router v7

## Backend

Supabase Cloud (PostgreSQL), Supabase JS Client

## UI Components

Lucide React Icons

## Hosting

Firebase Hosting

# Role-Based Access Control

## LIBRARIAN (Admin)

- Manage books (CRUD)
- Manage members (CRUD)
- Issue & process returns
- Manage authors & categories
- Generate reports & export CSV

## READER (Member)

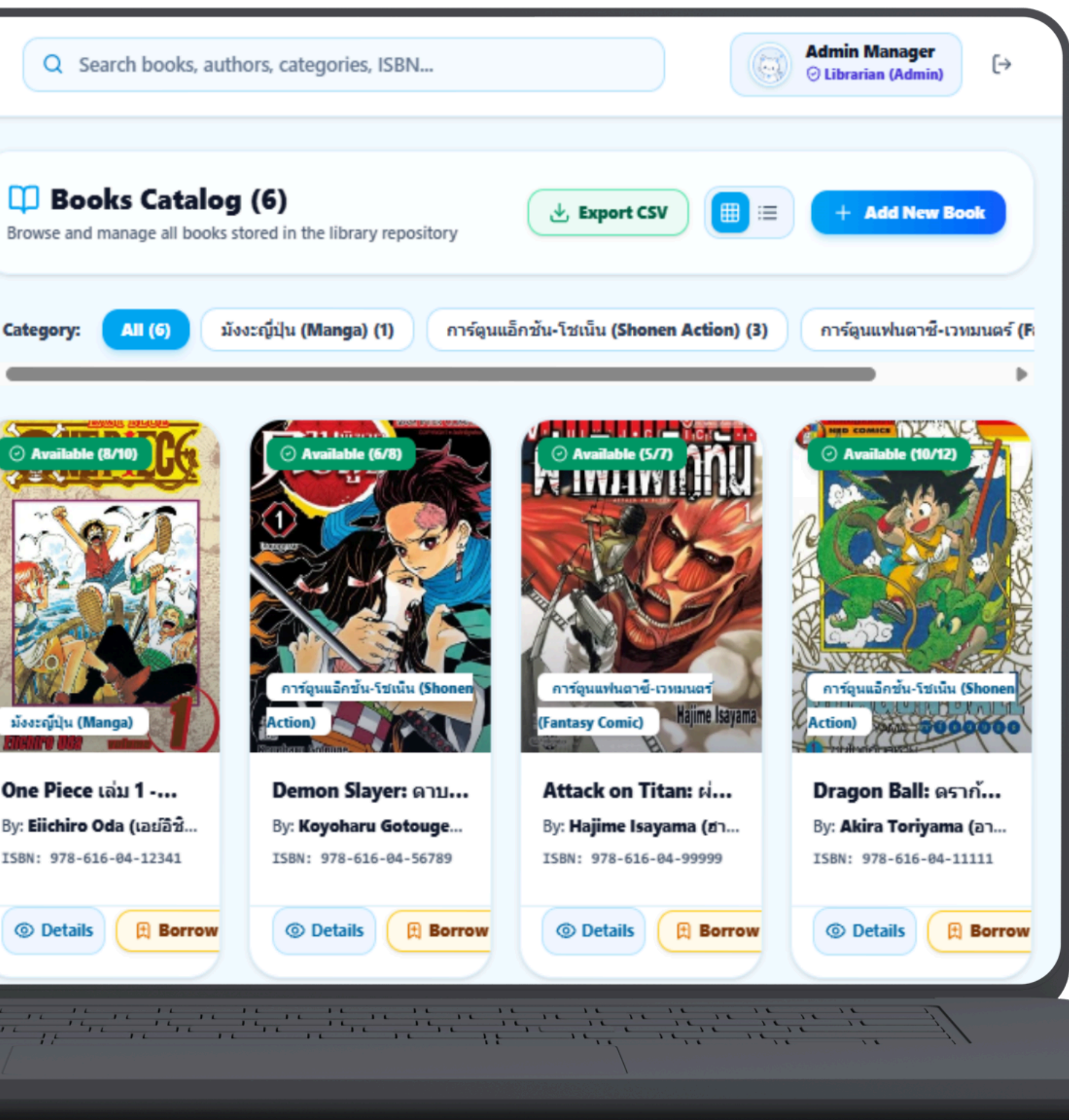
- Search & browse books
- View categories
- Check personal borrowing history

# Demo Credentials

| Role          | Email               | Password | Access    |
|---------------|---------------------|----------|-----------|
| Librarian     | admin@nekoshelf.com | 123456   | Full      |
| Reader/Member | user@nekoshelf.com  | 123456   | Read-Only |

# Key Achievements

- Full-stack web application with React & Supabase
- Relational database with 6 normalized tables
- Role-based access control (Admin & Member)
- Real-time inventory tracking & automated calculations
- CSV export for reporting & data analysis
- Deployed on Firebase Hosting with production readiness



A Modern Library Management System

TRY!



<https://nekoshelf.web.app/>